ELEC3705 Assignment

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1 Introduction

In this assignment, we simulate Nuclear Magnetic Resonance (NMR) imaging of a two-dimensional grayscale image, where the value of each pixel is the number of hydrogen nuclei at that location. A strong static field B_z, whose magnitude depends on the column index, is combined with a uniform oscillating field $B_x(t)$ at a fixed frequency. Only columns whose Larmor frequency is close to the drive frequency are strongly excited. By moving the centre of the field profile across the image and recording the total signal of each row, the image can be reconstructed column by column.

The nuclei do not interact with each other, so each pixel is modelled as one isolated spin-$\frac{1}{2}$, with the number of nuclei in the pixel acting as a multiplicative factor. The Hamiltonian is $\mathcal{H} = \gamma_n \mathbf{B}\cdot\mathbf{I}$, with $\mathbf{I} = \sigma/2$, and $\gamma_n = 42.48\,\text{MHz/T}$, and it is expressed in Hz. The Schrödinger equation is then

$$i\frac{d\,|\psi\rangle}{dt} = 2\pi\mathcal{H}\,|\psi\rangle\,,$$

so for a time-independent Hamiltonian the time evolution operator is

$$U(t) = e^{-i2\pi\mathcal{H}t}. \tag{1}$$

The factor of 2π is the conversion $h/\hbar$.

2 Question 1.1: Image Size

After loading the image, the shape of the array is 255×255. Note that $c = 0$ corresponds to the centre pixel of the image, which in this case is 127.

3 Question 1.2: Zeeman Energies

The static field is a Gaussian dip in the column coordinate, given by

$$B_z(c) = 1 - \frac{G(c)}{2\max G(c)}\ \text{T}, \tag{2}$$

$$G(c) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(c-\mu)^2/2\sigma^2}, \quad \sigma = 10. \tag{3}$$

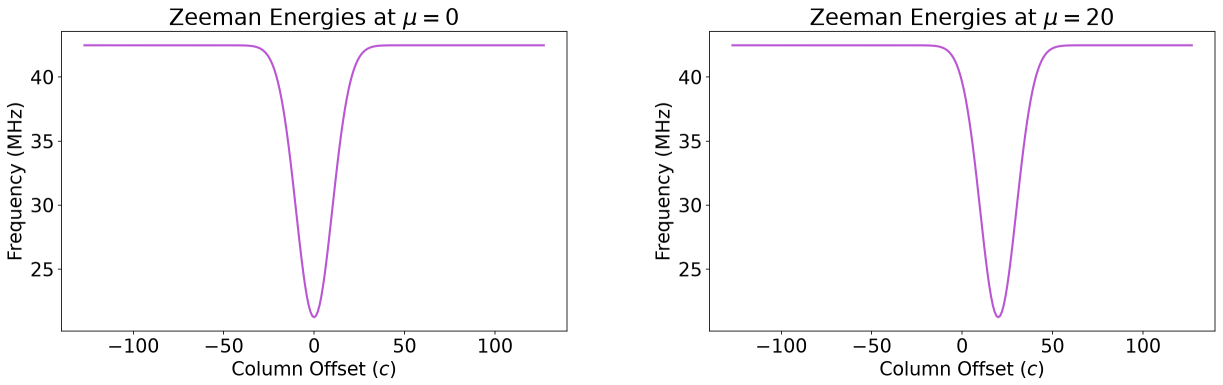
Since $\max G = G(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$, the ratio is $\frac{G}{2\max G} = \frac{1}{2}e^{-(c-\mu)^2/2\sigma^2}$, so $B_z = 1$ T far from μ, and $B_z = 0.5$ T at $c = \mu$. The Hamiltonian in this field is

$$\mathcal{H} = \gamma B_z I_z \quad I_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{4}$$

which is diagonal with eigenvalues $\lambda = \pm\gamma_n B_z/2$. The two spin levels are therefore split by

$$E_z = \gamma_n B_z, \tag{5}$$

which is the Larmor frequency. Fig. 1 shows $E_z(c)$ for the field centred at the middle pixel and shifted 20 pixels to the right. It is an inverted Gaussian, falling from 42.48 MHz far from the centre to 21.24 MHz at $c = \mu$.



(a) Field centred at column 127 ($\mu = 0$).

(b) Field centred at column 147 ($\mu = 20$).

Figure 1: Zeeman energy $E_z = \gamma_n B_z(c)$ (the splitting between the two spin levels) as a function of column offset c.

4 Question 1.3: Hamiltonian of Each Pixel

With $B_x = B_y = 0$, only the I_z term of $\gamma_n \mathbf{B} \cdot \mathbf{I}$ has an effect. Substituting Eq. (2) into Eq. (4) and centring the field on the middle pixel ($\mu = 0$),

$$\mathcal{H}(c) = \gamma_n \left(1 - \frac{1}{2} e^{-c^2/200}\right) I_z. \tag{6}$$

The Hamiltonian is diagonal because there is no transverse field yet. For the central column ($c = 0$),

$$\mathcal{H}(0) = \frac{1}{2}\gamma I_z = 21.24\,\text{MHz} \cdot I_z = \begin{pmatrix} 10.62 & 0 \\ 0 & -10.62 \end{pmatrix} \text{MHz}, \tag{7}$$

and for the last column ($c = 127$), where $e^{-127^2/200} \approx 10^{-35}$ is negligible,

$$\mathcal{H}(127) \approx \gamma_n I_z = 42.48\,\text{MHz} \cdot I_z = \begin{pmatrix} 21.24 & 0 \\ 0 & -21.24 \end{pmatrix} \text{MHz}. \tag{8}$$

The diagonal entries are half the Larmor frequency because I_z has eigenvalues $\pm\frac{1}{2}$. Every pixel in a column shares the same Hamiltonian, so we store one 2×2 matrix per column.

5 Question 1.4: Rabi Frequency and the Rotating Frame

We now add the oscillating field $B_x(t) = B_{amp} \cos(2\pi f t)$ with $B_{amp} = 50\text{mT}$. The laboratory-frame Hamiltonian of column c is given by

$$\mathcal{H}(t) = f_L(c) I_z + \gamma_n B_{amp} \cos(2\pi f t) I_x, \quad f_L(c) = \gamma_n B_z(c). \tag{9}$$

A linearly oscillating field is the sum of two fields rotating in opposite directions about z,

$$B_x = B_{amp} \frac{e^{i2\pi f t} + e^{-i2\pi f t}}{2}, \tag{10}$$

each of amplitude $\frac{B_{amp}}{2}$. The component $e^{i2\pi f t}$ rotates with the precessing spins and drives them at resonance. The component $e^{-i2\pi f t}$ is detuned by $2f$, and so has not effect (this is rotating-wave approximation).

The drive frequency was chosen to be equal to the Larmor frequency of the central column,

$$f = f_L(0) = \gamma_n(0.5\text{ T}) = 21.24\,\text{MHz}. \tag{11}$$

Transforming to the frame rotating about z at frequency f, $\left|\tilde{\psi}\right\rangle = R(t)\left|\psi\right\rangle$ with $R(t) = e^{i2\pi f t I_z}$, gives

$$i\frac{d\left|\tilde{\psi}\right\rangle}{dt} = 2\pi\tilde{\mathcal{H}}\left|\psi\right\rangle, \tag{12}$$

with $\tilde{\mathcal{H}} = RHR^\dagger - fI_z$. Using $RI_x r^\dagger = I_x\cos(2\pi ft) - I_y\sin(2\pi ft)$,

$$\tilde{\mathcal{H}} = (f_L - f)\,I_z + \gamma_n B_{amp}\left[\frac{1}{2}I_x\left(1+\cos(4\pi ft)\right) - \frac{1}{2}I_y\sin(2\pi ft)\right]. \tag{13}$$

The terms oscillating at $2f = 42.48\,\text{MHz}$ are much faster than the dynamics we resolve ($\sim_1$ MHz) and average out, dropping them is the rotating-wave approximation. The constant $\frac{1}{2}I_x$ term is the co-rotating half of the drive, and it is where the factor of $\frac{1}{2}$ from below comes from. The rotating frame Hamiltonian is then

$$\tilde{\mathcal{H}(c)} = \Delta(c)I_z + \Omega_R I_x \quad \Delta(c) = f_L(c) - f, \quad \Omega_R = \frac{\gamma_n B_{amp}}{2}. \tag{14}$$

For the central column $\Delta = 0$, so $\tilde{\mathcal{H}} = \Omega_R I_x$ and the spin rotates about x at angular rate $2\pi\Omega_R$. The Rabi frequency and period are then

$$\Omega_R = \frac{(42.48\,\text{MHz/T})(0.05\text{mT})}{2} = 1.062\,\text{MHz}, \quad T_{\text{Rabi}} = \frac{1}{\Omega_R} \approx_0 .94\,\mu\text{s}. \tag{15}$$

We can then construct our Hamiltonians. For the central column ($\Delta = 0$)

$$\tilde{\mathcal{H}}(0) = \Omega_R I_x = \begin{pmatrix} 0 & 0.531 \\ 0.531 & 0 \end{pmatrix}\text{MHz}, \tag{16}$$

and for the last column ($\Delta = 42.48 - 21.24 = 21.24\,\text{MHz}$)

$$\tilde{\mathcal{H}}(127) = 21.24\,\text{MHz}I_z + 1.062\,\text{MHz}I_x = \begin{pmatrix} 10.62 & 0.531 \\ 0.531 & -10.62 \end{pmatrix}\text{MHz}. \tag{17}$$

6 Question 1.5: Initial State

All spins are initialised in the down state $|\Downarrow\rangle = (0,1)^T$. Since this is an eigenstate of I_z with eigenvalue $-\frac{1}{2}$, and does not depend on $B_z(c)$, we expect $\langle I_z\rangle = -\frac{1}{2}$ at every position, which is what Fig. 2 shows. The column-dependent field changes each spin's energy, not its state, so the position dependence only appears once the spins evolve.

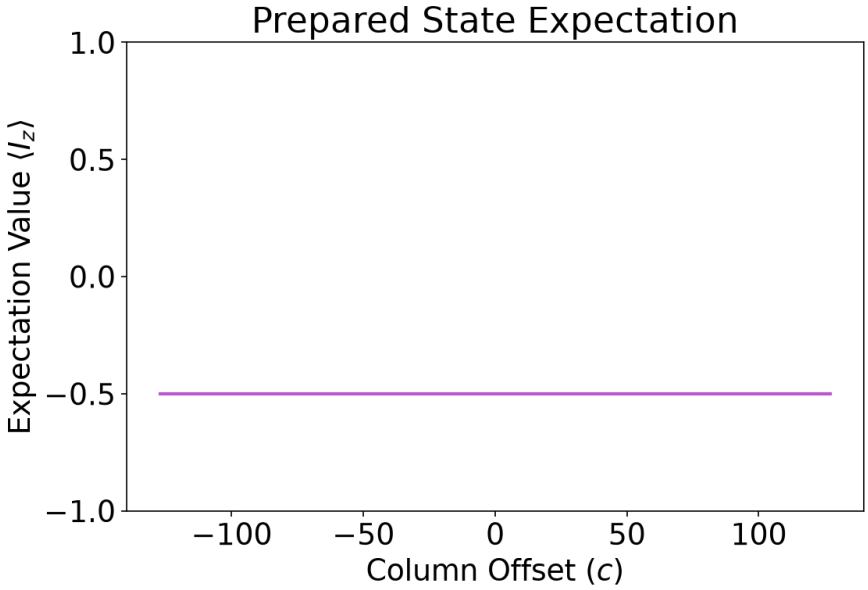


Figure 2: $\langle I_z \rangle$ for each pixel in a row when all spins start in $|\Downarrow\rangle$. All rows are identical, so this row is representative of every row.

7 Question 1.6

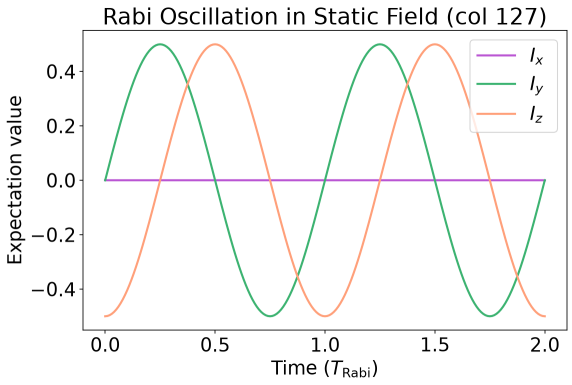
Starting from $|\Downarrow\rangle$, each column is evolved under its rotating-frame Hamiltonian $\tilde{\mathcal{H}}(c)$ of Eq. (14), which is time-independent in the rotating frame after the RWA, so $|\tilde{\psi}(t)\rangle = e^{-i2\pi\tilde{\mathcal{H}}t}\,|\Downarrow\rangle$. The time series covers $2T_{\mathrm{Rabi}}$ in steps of $T_{\mathrm{Rabi}}/100$ (201 points). The results for the central column and for columns offset by 5, 10 and 127 are in Fig. 3.

A spin detuned by Δ oscillates at the generalised Rabi frequency $\Omega_{\mathrm{eff}} = \sqrt{\Omega_R^2 + \Delta^2}$, with contrast $\Omega_R^2/\Omega_{\mathrm{eff}}^2$: $\langle I_z \rangle$ rises from $-\frac{1}{2}$ by at most that amount. Table 1 lists these values and matches the plots.

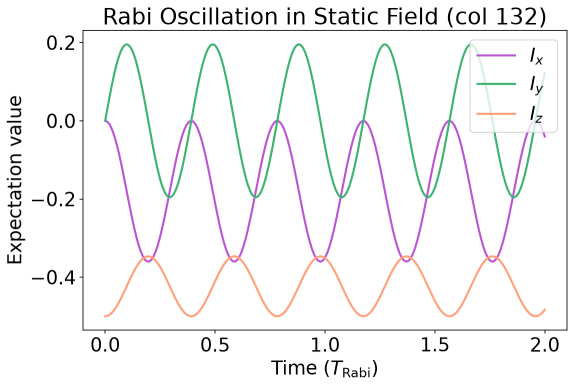
Offset c	Δ (MHz)	Ω_{eff} (MHz)	Contrast	Cycles in $2T_{\mathrm{Rabi}}$
0	0	1.06	100%	2
5	2.50	2.71	15%	5.1
10	8.36	8.43	1.6%	15.9
127	21.24	21.27	0.25%	40.1

Table 1: Detuning, generalised Rabi frequency, contrast and number of oscillations in $2T_{\mathrm{Rabi}}$ for the columns in Fig. 3.

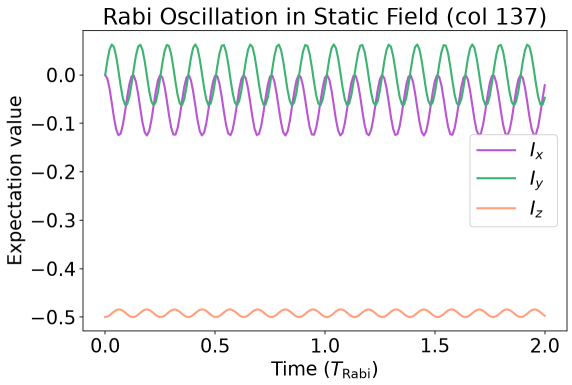
In the central column the spin is flipped completely, with $\langle I_z \rangle$ swinging from $-\frac{1}{2}$ to $+\frac{1}{2}$ twice and $\langle I_x \rangle = 0$. As the detuning grows the oscillation becomes faster and its contrast quickly shrinks, so a column only a few pixels from the centre is already driven weakly, and the last column hardly moves from $|\Downarrow\rangle$. This is the column selectivity that the reconstruction in Question 1.10 relies on.



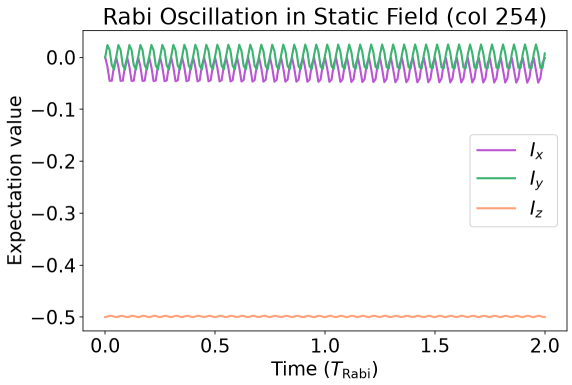
(a) Central column (column 127, $c = 0$): on resonance, full contrast.



(b) Column 132 ($c = 5$).



(c) Column 137 ($c = 10$).



(d) Last column (column 254, $c = 127$): barely driven.

Figure 3: Expectation values $\langle I_x \rangle$, $\langle I_y \rangle$ and $\langle I_z \rangle$ under the rotating-frame Hamiltonian $\tilde{\mathcal{H}}(c)$ for $2T_{\mathrm{Rabi}}$.

8 Question 1.7: $\pi/2$ Pulse

On resonance the pulse is a rotation about x by the angle $\theta = 2\pi\Omega_R t$. A $\pi/2$ pulse takes the spin from $|\Downarrow\rangle$ into the xy-plane, so $\theta = \pi/2$ and

$$t_{\pi/2} = \frac{1}{4\Omega_R} = \frac{T_{\text{Rabi}}}{4} \approx 0.235\,\mu\text{s}. \tag{18}$$

This matches Question 1.6, in Fig. 3(a), $\langle I_z \rangle$ crosses zero at $0.25\,T_{\text{Rabi}}$. Since the spins start at $t = 0$, the state after the pulse is simply

$$|\tilde{\psi}(t_{\pi/2})\rangle = U(t_{\pi/2})\,|\Downarrow\rangle\,, \qquad U(t_{\pi/2}) = e^{-i2\pi\tilde{\mathcal{H}}t_{\pi/2}}, \tag{19}$$

and no new time evolution is needed. The result is shown in Fig. 4. In the central column $\langle I_y \rangle = +\frac{1}{2}$ and $\langle I_x \rangle = \langle I_z \rangle = 0$, i.e. the spin lies along y. The neighbouring columns show ringing from their different detunings, and the far columns stay essentially in $|\Downarrow\rangle$.

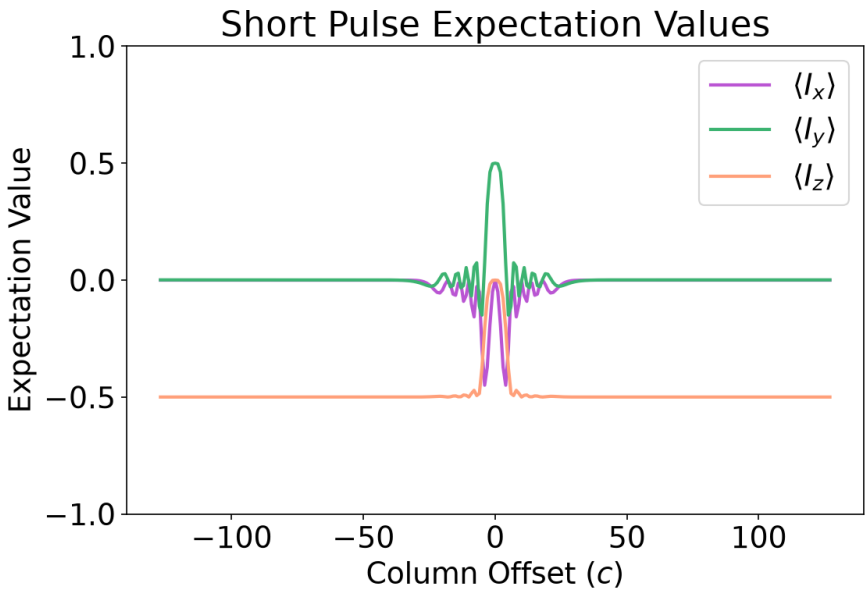
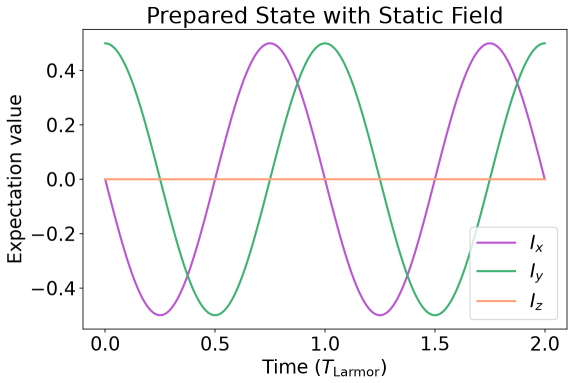


Figure 4: Expectation values of each column after a $\pi/2$ pulse. Only columns near resonance (the centre) are tipped into the xy-plane.

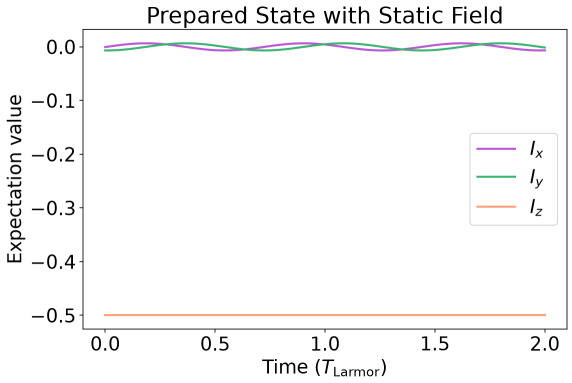
9 Question 1.8: Free Precession

We now switch the oscillating field off, so each spin evolves under the laboratory-frame Hamiltonian of Question 1.3, $\mathcal{H}(c) = \gamma_n B_z(c) I_z$, starting from the states of Question 1.7. We simulate $2T_{\text{Larmor}}$ with $T_{\text{Larmor}} = 1/(21.24\,\text{MHz}) \approx 47.1\,\text{ns}$ (the Larmor period of the central spin), in steps of $T_{\text{Larmor}}/100$. The results are in Fig. 5.

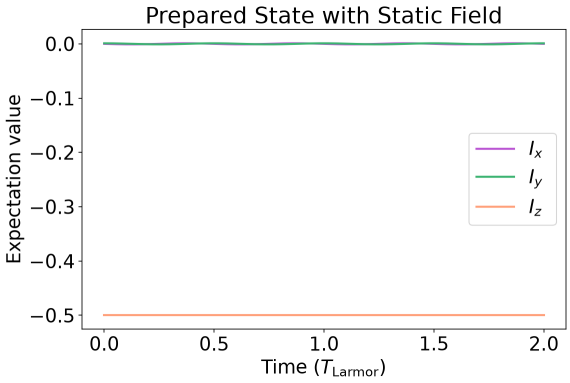
Each spin precesses about z at its own Larmor frequency $f_L(c) = \gamma_n B_z(c)$: 21.24, 29.60 and 42.48 MHz for the central column, the column offset by 10, and the last column. This gives 2, 2.8 and 4 cycles in the window. The central column was tipped fully into the xy-plane, so $\langle I_x \rangle$ and $\langle I_y \rangle$ oscillate with amplitude $\frac{1}{2}$ as a quadrature pair while $\langle I_z \rangle = 0$. The other two columns were hardly tipped by the pulse, so $\langle I_z \rangle \approx -\frac{1}{2}$ and their transverse component is small and nearly invisible at this scale. Every spin precesses; what falls with distance from resonance is the transverse amplitude left by the $\pi/2$ pulse. Because the precession frequency depends on column, the signal of a spin encodes its position, which the readout in Questions 1.9 and 1.10 uses.



(a) Central column ($c = 0$).



(b) Column offset by 10 ($c = 10$).



(c) Last column ($c = 127$).

Figure 5: Larmor precession of different columns in the static field after the $\pi/2$ pulse, for $2T_{\text{Larmor}}$.

10 Question 1.9: Aggregated Signal

Each pixel holds many nuclei, so the detector signal of row r is the sum over columns of the number of spins $N(r, c)$ in each pixel times that pixel's $\langle I_x \rangle$,

$$P(r, t) = \sum_c N(r, c)\, \langle I_x \rangle_c(t). \tag{20}$$

Fig. 6 shows $P(t)$ for the central row ($r = 127$). The signal is dominated by the columns near resonance, which were tipped into the xy-plane. It is not a clean sinusoid, because neighbouring columns precess at different frequencies and dephase, and the weights $N(r, c)$ differ between pixels. A single readout cannot resolve individual columns, since the detector only measures the sum.

The maximum of $P(t)$ occurs at $t_{\max} = 0.65\, T_{\text{Larmor}} \approx 30.6\,\text{ns}$, which we use as the readout time in Question 1.10.

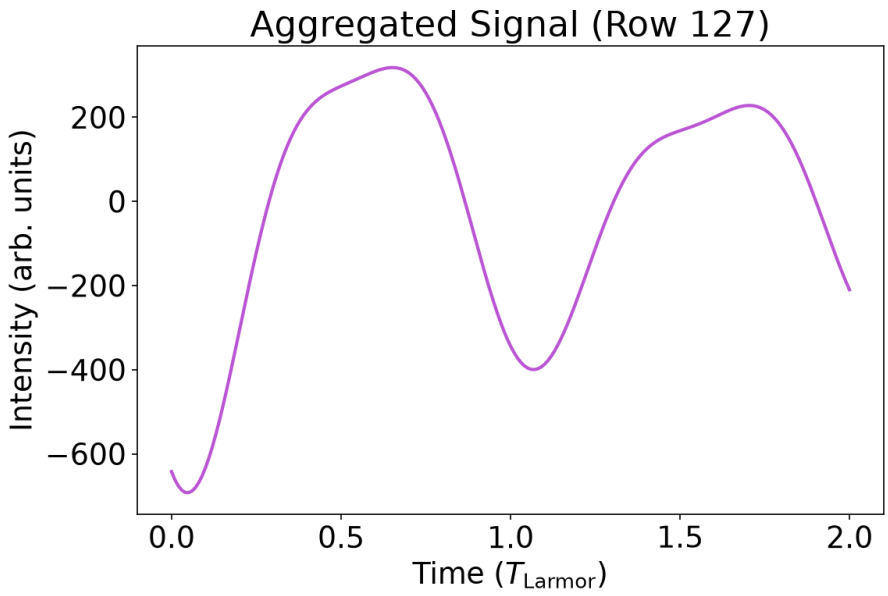


Figure 6: Aggregated signal $P(t)$ of row 127 over $2T_{\text{Larmor}}$. The maximum, at $t_{\max} = 0.65\, T_{\text{Larmor}}$, is the readout time used for the reconstruction.

11 Question 1.10: Reconstruction

The field profile is shifted by changing the parameter μ in $G(c)$, which sets the column at which $B_z = 0.5\text{T}$, i.e. the column resonant with the fixed drive frequency $f = 21.24\,\text{MHz}$. For each μ (each column of the image) , we recompute $B_z(c)$ and $\tilde{\mathcal{H}}(c)$ for all columns since the detunings $\Delta(c)$ change with μ, apply the $\pi/2$ pulse in the rotating frame starting from $|\Downarrow\rangle$, precess in the laboratory frame for $t_{\max}$ (the same value works for every μ because the resonant column always has the same Larmor frequency), and compute $\langle I_x \rangle_c$ for each column and form the weighted sum over each row

$$I_{\text{rec}}(r, \mu) = \sum_c N(r, c) \langle I_x \rangle_c(t_{\max}, \mu), \tag{21}$$

which gives one column of the reconstructed image. The result is shown in Fig. 7

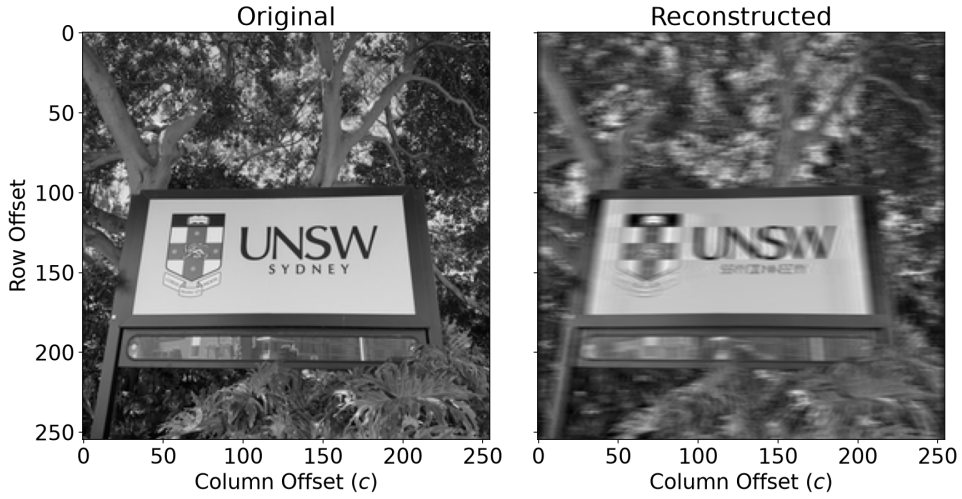


Figure 7: Original image (left) and NMR reconstruction (right).

The reconstruction is recognisable as the original image, so scanning the field centre does recover the image, and it makes sense physically. The position is encoded in $B_z(c)$, only spins near the resonance condition respond strongly to the pulse, and moving the resonance across the image reads it out column by column. It differs from the original in three ways.

- **Horizontal blurring.** The excitation is not perfectly selective. Columns within a few pixels of μ are also partly tipped (Table 1), so they contribute to the signal of the column being read. The width of the resonance, set by Ω_R relative to how quickly B_z changes, limits the horizontal resolution.
- **Ringing and sign changes.** Neighbouring columns precess at different frequencies, so by $t_{\max}$ they have accumulated phase differences of order a radian or more (about 1.6 rad for a column detuned by 8.4 MHz). Their contributions have different signs and partly cancel, which produces ringing near sharp vertical edges.
- **Different intensity scale.** $\langle I_x \rangle \leq \frac{1}{2}$, and partially tipped or out-of-phase neighbours reduce it further, so the absolute scale is not the original's.

The image is not blurred vertically because B_z does not depend on the row, and so every row sees the same fields and is read out independently.

12 Conclusion

We modelled the nuclei as non-interacting spin-$\frac{1}{2}$ particles in a column-dependent static field and a uniform oscillating drive. In the rotating frame the drive gives a Rabi frequency $\Omega_R = \gamma_n B_{amp}/2 = 1.09\,\text{MHz}$, and only columns close to resonance are driven strongly. A $\pi/2$ pulse followed by free precession encodes each column in its Larmor frequency, and scanning the field centre across the image reconstructs it, up to a horizontal blur and ringing set by the finite width of the resonance.